Proof. We drop the superscripts i, l in the proof to avoid notation clutter. Similarly, we denote the concatenation of the prompt matrix $\mathbf{P}$ and the original input matrix $\mathbf{X}$, simply with $\mathbf{X}$.

Derivation for single head eq 18:
Consider two matrices $\mathbf{X}_1, \mathbf{X}_2 \in \mathcal{X} = \{\mathbf{X} \in \mathbb{R}^{d\times m}; \|\mathbf{X}\|_2 \leq D\}$. Denote with $\mathbf{A}_1, \mathbf{A}_2$ the corresponding attention matrices respectively, which can be defined as:

$$\begin{aligned} \mathbf{A}_1 &= \sigma((\mathbf{W}_k\mathbf{X}_1)^\top \mathbf{W}_q\mathbf{X}_1) \\ \mathbf{A}_2 &= \sigma((\mathbf{W}_k\mathbf{X}_2)^\top \mathbf{W}_q\mathbf{X}_2) \end{aligned} \tag{20}$$

The output of the attention head, denoted with $Att(\cdot)$ admits the following:

$$\begin{aligned} \|Att(\mathbf{X}_1) - Att(\mathbf{X}_2)\|_2 &= \|\mathbf{W}_v\mathbf{X}_1\mathbf{A}_1 - \mathbf{W}_v\mathbf{X}_2\mathbf{A}_2\|_2 && (21) \\ &\overset{a}{\leq} \|\mathbf{X}_1\mathbf{A}_1 - \mathbf{X}_2\mathbf{A}_2\|_2\|\mathbf{W}_v\|_2 && (22) \\ &= \|\mathbf{X}_1\mathbf{A}_1 - \mathbf{X}_2\mathbf{A}_1 + \mathbf{X}_2\mathbf{A}_1 - \mathbf{X}_2\mathbf{A}_2\|_2\|\mathbf{W}_v\|_2 && (23) \\ &\leq (\|\mathbf{A}_1\|_2\|\mathbf{X}_1 - \mathbf{X}_2\|_2 + \|\mathbf{X}_2\|_2\|\mathbf{A}_1 - \mathbf{A}_2\|_2)\|\mathbf{W}_v\|_2 && (24) \\ &\overset{b}{\leq} (\|\mathbf{X}_1 - \mathbf{X}_2\|_2 + \|\mathbf{A}_1 - \mathbf{A}_2\|_2 D)\|\mathbf{W}_v\|_2 && (25) \end{aligned}$$

where (a) holds from the spectral norm properties and in (b) we use the bounded input spectral norm assumptions.
We now focus on the second term $\|\mathbf{A}_1 - \mathbf{A}_2\|_2$ in eq 25 . From the bound in lemma 9, we have:

$$\|\mathbf{A}_1 - \mathbf{A}_2\|_2 \leq 2\sqrt{m}\|\mathbf{G}\|_2 \tag{26}$$

where $\mathbf{G}$ is the diagonal matrix with entires described in lemma 8
We can now invoke lemma 10 to obtain the following :

$$\|\mathbf{A}_1 - \mathbf{A}_2\|_2 \leq 2\sqrt{m} \times 2 \times 2\|\mathbf{W}_k^T\mathbf{W}_q\|_2 D \times \|\mathbf{X}_1 - \mathbf{X}_2\|_2 \tag{27}$$

Combining the previous inequality with eq 25, we have the following bound:

$$\|Att(\mathbf{X}_1) - Att(\mathbf{X}_2)\|_2 \leq (1 + 8\sqrt{m}\|\mathbf{W}_k^T\mathbf{W}_q\|_2 D^2)\|\mathbf{W}_v\|_2\|\mathbf{X}_1 - \mathbf{X}_2\|_2 \tag{28}$$

Derivation for the entire block eq 11:
The proof follows simply by leveraging the following property:
Property: for a matrix $\mathbf{C} = [\mathbf{A}, \mathbf{B}]$, the spectral norm of $\mathbf{C}$ admits the bound:

$$\|\mathbf{C}\|_2 \leq \sqrt{\|\mathbf{A}\|_2^2 + \|\mathbf{B}\|_2^2}$$

We then simply combine the definition of the attention block and the lipschitz constant bound in eq 18 with the above property in order to obtain the desired bound. ☐

Lemma 8 (Dong et al. [2021] Lemma A.1)**.** *For the column stochastic matrix $\mathbf{A}_1$ obtained by performing column-wise softmax of some matrix $\mathbf{Z}_1$ (where in our setting $\mathbf{Z}_1 = (\mathbf{W}_k\mathbf{X}_1)^\top\mathbf{W}_q\mathbf{X}_1$, and another row stochastic matrix $\mathbf{A}_2$ obtained by performing column-wise softmax of some matrix $\mathbf{Z}_2$, where $\mathbf{Z}_2 = \mathbf{Z}_1 - \mathbf{E}$ (for some $\mathbf{E}$, which **need not** belong to $\mathcal{X}$), we have the following bound:*

$$\mathbf{A}_2(\mathbf{I} - \mathbf{G}) \leq \mathbf{A}_1 \leq \mathbf{A}_2(\mathbf{I} + 2\mathbf{G}) \tag{29}$$

where the inequality is elementwise and $\mathbf{G}$ is a diagonal matrix with entries as $\mathbf{G}_{ii} = \max_{j,j'} |\delta_i^T\mathbf{E}(\delta_j^T - \delta_{j'}^T)|$. Here δ_k is a one-hot vector with the entry 1 in the k^{th} dimension.

Lemma 9. *Following the notations of lemma 8, we have the following spectral norm bound:*

$$\|\mathbf{A}_1 - \mathbf{A}_2\|_2 \leq 2\sqrt{m}\|\mathbf{G}\|_2 \tag{30}$$

Proof. We begin by noting the following entry-wise inequality from eq 29:

$$\mathbf{A}_2\mathbf{G} \leq \mathbf{A}_1 - \mathbf{A}_2 \leq 2\mathbf{A}_2\mathbf{G} \tag{31}$$

which ensures that $\|\mathbf{A}_1 - \mathbf{A}_2\|_F \leq 2\|\mathbf{A}_2\mathbf{G}\|_F$.
We also have the following using matrix norm equivalence:

$$\|\mathbf{A}_1 - \mathbf{A}_2\|_2 \leq \|\mathbf{A}_1 - \mathbf{A}_2\|_F \tag{32}$$

Invoking the matrix norm equivalence again, we have that

$$2\|\mathbf{A}_2\mathbf{G}\|_F \leq 2\sqrt{rank(\mathbf{A}_2\mathbf{G})}\|\mathbf{A}_2\mathbf{G}\|_2 \tag{33}$$

where $rank(\cdot)$ is the matrix rank.
Combining the inequalities, we attain the bound :

$$\|\mathbf{A}_1 - \mathbf{A}_2\|_2 \leq 2\sqrt{m}\|\mathbf{A}_2\|_2\|\mathbf{G}\|_2 \tag{34}$$

since $\mathbf{A}_2$ is column-stochastic , $\|\mathbf{A}_2\|_2 = 1$ $\square$

Lemma 10. *The term $\mathbf{G}$ in lemma 9 admits the following spectral norm bound:*

$$\|\mathbf{G}\|_2 \leq 2D\|\mathbf{W}_q\mathbf{W}_k^T\|_2\|\mathbf{X}_1 - \mathbf{X}_2\|_2 \tag{35}$$

here D is the previously stated spectral norm bound of the inputs $\mathbf{X}^l \in \mathcal{X}^l$.

Proof. We begin by noting that since $\mathbf{G}$ is a square diagonal matrix with non-negative real values, the singular values of $\mathbf{G}$ are the corresponding diagonal elements.
We thus have that $\|\mathbf{G}\|_{max} = \|\mathbf{G}\|_2$, where $\|\cdot\|_{max}$ is the max norm.
Since $\mathbf{G}$ admits the form described in lemma 8, it is trivial to note that:

$$\|\mathbf{G}\|_{max} = \max_{i,j,i',j'} |\mathbf{E}_{i,j} - \mathbf{E}_{i',j'}| \tag{36}$$

$$\leq 2\|\mathbf{E}\|_{max} \leq 2\|\mathbf{E}\|_2 \tag{37}$$

where the second inequality follows from the matrix norm equivalence.
Now, we can bound the last term $\|\mathbf{E}\|_2$ by noting that the inputs $\mathbf{X}$ belong to a bounded set. This allows us to provide the following bounds:

$$\|\mathbf{E}\|_2 = \|(\mathbf{W}_k\mathbf{X}_1)^\top\mathbf{W}_q\mathbf{X}_1 - (\mathbf{W}_k\mathbf{X}_2)^\top\mathbf{W}_q\mathbf{X}_2\|_2 \tag{38}$$

$$= \|(\mathbf{W}_k\mathbf{X}_1)^\top\mathbf{W}_q\mathbf{X}_1 - (\mathbf{W}_k\mathbf{X}_1)^\top\mathbf{W}_q\mathbf{X}_2 + (\mathbf{W}_k\mathbf{X}_1)^\top\mathbf{W}_q\mathbf{X}_2 - (\mathbf{W}_k\mathbf{X}_2)^\top\mathbf{W}_q\mathbf{X}_2\|_2 \tag{39}$$

$$\leq (\|\mathbf{X}_1\|_2\|\mathbf{W}_k^T\mathbf{W}_q\|_2 + \|\mathbf{X}_2\|_2\|\mathbf{W}_k^T\mathbf{W}_q\|_2)\|\mathbf{X}_1 - \mathbf{X}_2\|_2 \tag{40}$$

$$\leq 2D\|\mathbf{W}_k^T\mathbf{W}_q\|_2\|\mathbf{X}_1 - \mathbf{X}_2\|_2 \tag{41}$$

$\square$

C.11 Extension of Lemma 6

Lemma 6 and theorem 4 operate over functions from $\mathcal{P}^1 \times \mathcal{X}^1 \to \mathcal{P}^{L+1} \times \mathcal{X}^{L+1}$. We can relax the requirement of the prompt and provide the Lipschitz constant upper bound in consideration to functions of the form $\mathcal{X}^1 \to \mathcal{X}^{L+1}$ by using the following assumption:

Assumption 3. *Assume for simplicity that $\|\mathbf{P}_1^l - \mathbf{P}_2^l\|_2 \leq \alpha^l\|\mathbf{X}_1^l - \mathbf{X}_2^l\|_2; \forall l \geq 1$. $\alpha^l = 0$ when $l = 1$.*

Note: A recursive expression for α^l in the above assumption can be provided, but the expression does not admit a simplified form and we thus omit it here.

We will use $\mathcal{D}_X^l$, akin to eq 9, to denote the compactness corresponding to the input matrix across the layers.
Based on this assumption, we have the following Lipschitz constant upper bound:

Lemma 11. *The Lipschitz constant of the single head $\mathtt{Att}^{i,l}$ admits the following bound w.r.t the input part, $\mathbf{X}^1$ of length m_X, of the input sequence:*

$$Lip(\mathtt{Att}^{i,l}(\cdot,\cdot)) \leq \left(\sqrt{1 + (\alpha^l)^2} + 8\sqrt{m_X}(D_X^l)^2(1 + (\alpha^l)^2)\|(\mathbf{W}_k^{i,l})^T\mathbf{W}_q^{i,l}\|_2\right)\|\mathbf{W}_v^{i,l}\|_2 \tag{42}$$

For $l = 1$, $\alpha^l = 0$ in the above bound.
The Lipschitz constant of the entire attention block in layer l follows similarly.

Proof. For some first layer input $\mathbf{X}_1^1$ and prompt $\mathbf{P}$, let us denote the direct output of the attention head in the l-th layer with $\overrightarrow{\mathbf{X}}_1^l$. We have the following update rule for $\overrightarrow{\mathbf{X}}_1^l$:

$$\overrightarrow{\mathbf{X}}_1^l = \mathbf{W}_v^{i,l}[\mathbf{P}_1^l, \mathbf{X}_1^l] \cdot \sigma((\mathbf{W}_k^{i,l}[\mathbf{P}_1^l, \mathbf{X}_1^l])^\top \mathbf{W}_q^i \mathbf{X}_1^l) = \mathbf{W}_v^{i,l}[\mathbf{P}_1^l, \mathbf{X}_1^l] \cdot \mathbf{A}_1^l \tag{43}$$

Here, $\mathbf{P}_1^l$ is the updated prompt matrix w.r.t the input. For two different inputs $\mathbf{X}_1^1$ and $\mathbf{X}_2^1$ at the first layer, $\mathbf{P}_1^1 = \mathbf{P}_2^1 = \mathbf{P}$, since the prompt is same across all inputs. $\mathbf{A}_1^l$ is then simply the corresponding column-stochastic matrix.

With the context clear, we now drop the superscripts i, l, as done previously. For $\|\overrightarrow{\mathbf{X}}_1 - \overrightarrow{\mathbf{X}}_2\|_2$, we have:

$$\begin{aligned} \|\overrightarrow{\mathbf{X}}_1 - \overrightarrow{\mathbf{X}}_2\|_2 &\leq \|[\mathbf{P}_1, \mathbf{X}_1]\mathbf{A}_1 - [\mathbf{P}_2, \mathbf{X}_2]\mathbf{A}_2\|_2 \|\mathbf{W}_v\|_2 \\ &\leq \left(\|\mathbf{A}_1\|_2 \sqrt{\|\mathbf{P}_1 - \mathbf{P}_2\|_2^2 + \|\mathbf{X}_1 - \mathbf{X}_2\|_2^2} + \sqrt{\|\mathbf{P}_2\|_2^2 + \|\mathbf{X}_2\|_2^2} \|\mathbf{A}_1 - \mathbf{A}_2\|_2 \right) \|\mathbf{W}_v\|_2 \end{aligned} \tag{44}$$

where the second inequality is attained using the property of spectral norm of concatenated matrices.

We now consider the term $\|\mathbf{A}_1 - \mathbf{A}_2\|_2$. By invoking lemmas 9 and 10, we have that:

$$\begin{aligned} &\|\mathbf{A}_1 - \mathbf{A}_2\|_2 \leq 2\sqrt{m_X} \times 2\|\mathbf{E}\|_2 \\ \text{where } \ &\mathbf{E} = (\mathbf{W}_k[\mathbf{P}_1, \mathbf{X}_1])^\top \mathbf{W}_q \mathbf{X}_1 - (\mathbf{W}_k[\mathbf{P}_2, \mathbf{X}_2])^\top \mathbf{W}_q \mathbf{X}_2 \end{aligned} \tag{45}$$

Invoking assumption 3 , $\|\mathbf{E}\|_2$ can further be bounded as:

$$\|\mathbf{E}\|_2 \leq 2D_X \sqrt{1+\alpha^2} \|(\mathbf{W}_k)^T \mathbf{W}_q\|_2 \tag{46}$$

Finally, by combining the bound for $\|\mathbf{E}\|_2$ and assumption 3 with eq 44, we obtain:

$$\|\overrightarrow{\mathbf{X}}_1 - \overrightarrow{\mathbf{X}}_2\|_2 \tag{47}$$

$$\leq \left(\sqrt{1+\alpha^2} + D_X \sqrt{1+\alpha^2} \times 8\sqrt{m_X} D_X \sqrt{1+\alpha^2} \|(\mathbf{W}_k)^T \mathbf{W}_q\|_2 \right) \|\mathbf{W}_v\|_2 \|\mathbf{X}_1 - \mathbf{X}_2\|_2 \tag{48}$$

which provides us the desired bound. □

By setting $\alpha^l = 0$, the case when there is no prompt, we obtain a similar bound as lemma 6